

Comprehensive Analyses and Comparison of 1 kW Isolated DC–DC Converters for Bidirectional EV Charging Systems

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Abstract—Isolated dc–dc converters with galvanic isolation are commonly used in electric vehicle (EV) battery chargers. These converters interface between a dc voltage link, which is usually the output of a power factor correction stage, and an energy storage unit. CLLC and dual active bridge (DAB) dc–dc converters can achieve high power density, high-energy efficiency, wide gain range, galvanic isolation, and bidirectional power flow, and therefore, have potential applications as dc–dc converters for bidirectional EV charging systems. In this paper, full-bridge CLLC, half-bridge CLLC, full-bridge DAB, and half-bridge DAB dc–dc converters are evaluated and compared for their suitability for EV chargers. All the converters are designed with optimal soft-switching features. The operating principles, design methodologies, and design considerations are presented. Prototypes of the converters with power rating of 1 kW are designed and developed. The prototypes interface a 500 V dc link and a 200–420 V load, which is common for EV applications. The performances of the circuits are analyzed and a comprehensive comparison is conducted.

Index Terms—Battery charger, bidirectional, CLLC, dual active bridge (DAB), isolated dc–dc converter, resonant converter, zero current switching (ZCS), zero voltage switching (ZVS).

I. INTRODUCTION

CONCERNS about global climate change and fossil fuel depletion are promoting the growth of the electric vehicle (EV) market [1]. Researchers have studied various techniques to improve the efficiency and power density of EV chargers [2]–[4]. A typical EV charger consists of an ac–dc power factor correction (PFC) stage and a galvanically isolated dc–dc converter [5]. The PFC stage, such as an interleaved boost converter, interfaces between the ac grid and a dc voltage link to achieve input PFC and ac–dc conversion [6], [7]. The dc–dc stage connects between the dc voltage link and an energy storage system [8].

For unidirectional grid-to-vehicle (G2V) chargers, LLC converters are commonly selected as the dc–dc stages [9], [10]. Recent studies indicate that EVs can also be considered as distributed power sources to store and send power back to the grid [11], [12]. This vehicle-to-grid (V2G) concept allows EVs to provide voltage and frequency regulation to the grid, absorb excess electricity, and deliver it to the

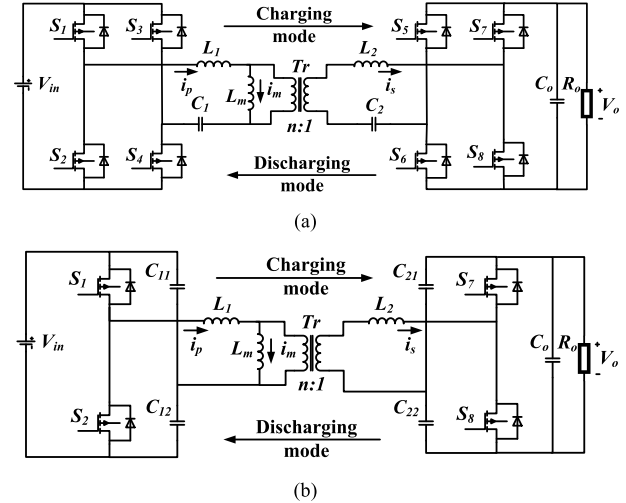


Fig. 1. Bidirectional (a) FBCLLC and (b) HBCLLC resonant converters.

grid during periods of high demand. The V2G capability requires EVs to have bidirectional charging systems that are able to operate in two power flow modes: charging mode (G2V) and discharging mode (V2G). The CLLC and dual active bridge (DAB) converters are commonly used as the dc–dc stages for bidirectional EV chargers, due to their advantages in terms of high power density, high efficiency, buck/boost capability, and controllable bidirectional power transfer [13]–[15].

Both CLLC and DAB circuits can be designed with full-bridge and half-bridge topologies. Different topologies require different design methods and are appropriate for different applications. A bidirectional full-bridge CLLC (FBCLLC) converter, illustrated in Fig. 1(a), is introduced in [16]. The converter is proposed for a 500 W power rating, 400 V input, and 48 V output UPS system, and possesses zero voltage switching (ZVS) and zero current switching (ZCS) features to minimize the switching loss. The highest efficiency of the prototype exceeds 96%. In [17], a design methodology for a 5 kW FBCLLC converter with soft start control is presented. The prototype is proposed for a 380 V dc power distribution system and the highest efficiency is 97.8% at 4 kW. A CLLC-compensated capacitive power transfer system for EV charging applications is proposed in [18]. The power rating of the system is 2.9 kW and the efficiency is 89.3%. The basic operating principles and simulation results of a half-bridge CLLC (HBCLLC) converter [shown in Fig. 1(b)] are reported in [19] without providing detailed experimental verification.

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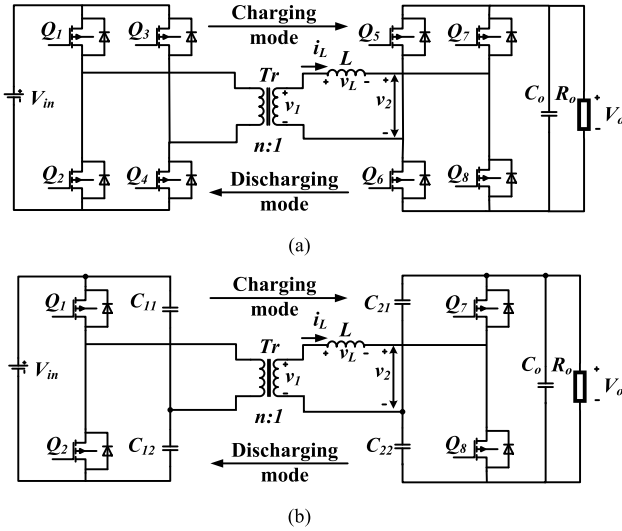


Fig. 2. Bidirectional (a) FBDAB and (b) HBDAB resonant converters.

Fig. 2(a) illustrates the topology of a full-bridge DAB (FBDAB) converter. The detailed operating principles, design considerations, and control method for a 10 kW FBDAB converter are discussed in [20]. Jeong *et al.* [21] present the design of a 5 kW prototype FBDAB converter for a 380 V input and 20–28 V output charger for a UPS system. The highest efficiency of this converter exceeds 96%. An analysis of the switching condition and loss modeling of GaN-based DAB converter for EV charger is provided in [22]; the modeling is verified by experimental measurement of a 500 kHz GaN DAB converter. A half-bridge DAB (HBDAB) converter [illustrated in Fig. 2(b)] for EV on-board charger is introduced in [23], and a 600 W prototype converter is built to verify the theoretical analysis and control method. By analyzing the optimal operation regions of both the FBDAB and HBDAB converters, [24] proposes a combined circuit that is able to switch between FBDAB and HBDAB to achieve high efficiency for wide loads. This is verified by an 800 W prototype and the maximum measured efficiencies are 92.9% at light load and 93.4% at full load.

Frequency modulation is the basic control method for both the FBCLLC and HBCLLC circuits. For FBDAB circuits, various control strategies, such as single-phase shift (SPS), extended-phase shift, dual-phase shift, and triple-phase shift controls can be applied through manipulating the ON/OFF status of the switches on both the primary and secondary sides of the converters [25]–[27]. For an HBDAB circuit, which consists of four switches, only the SPS control method can be applied. However, the half-bridge topology has benefits in terms of reduced size, weight, and cost, due to the reduction of the switches, corresponding driving circuits, and cooling systems. Therefore, it is necessary to discuss the advantages and disadvantages of the different topologies. In this paper, four dc–dc converter topologies, FBCLLC, HBCLLC, FBDAB, and HBDAB, are analyzed. The design methods of these four topologies for EV charging systems are introduced, and 1 kW prototypes for each topology are built to validate the analysis and designs. An important contribution of this paper is providing a general gain expression of the HBCLLC

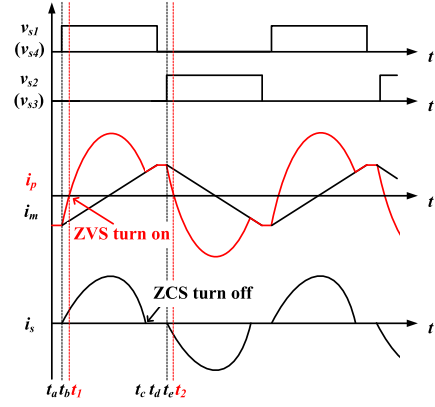


Fig. 3. Typical waveforms of a bidirectional FBCLLC or HBCLLC converter operating at a frequency lower than resonant frequency.

converter, and analyzing and verifying the practicability of the four isolated dc–dc converters for bidirectional EV charging systems.

This paper is organized as follows. The dc–dc converter topologies and their operating principles are discussed in Section II. Detailed design methodologies and considerations for EV charging systems are discussed in Section III. The experimental results are presented in Section IV and the analysis of the results is provided in Section V. Finally, Section VI provides the concluding remarks.

II. THEORETICAL ANALYSIS

CLLC and DAB converters are isolated switching converters. In order to ensure higher efficiency, the circuits need to be designed to operate under soft-switching conditions. In the case of CLLC converters, ZVS operation can be achieved on primary side power MOSFETs and ZCS operation can be realized on secondary side rectifiers. In the case of DAB converters, both primary and secondary side power MOSFETs can operate under ZVS condition.

A. Bidirectional CLLC Converters

1) *Operating Principles:* The bidirectional CLLC converters have symmetrical structures consisting of primary inverting stages and secondary rectifying stages. For the FBCLLC circuit [shown in Fig. 1(a)], L_1 and L_2 are resonant inductors and C_1 and C_2 are resonant capacitors. In comparison, the HBCLLC circuit [shown in Fig. 1(b)] uses bridge capacitors (C_{11} , C_{12} , C_{21} , and C_{22}) as resonant capacitors. The turns ratio of the transformer Tr is $n : 1$, and its magnetizing inductance is L_m . Fig. 3 illustrates typical waveforms of a bidirectional FBCLLC or HBCLLC circuit operating at a switching frequency lower than its resonant frequency. The dead band duration is the time between t_a and t_b , where all the switches are OFF to prevent the bridges from short-circuit shoot-through. In this interval, no power transfers to the secondary side, and the secondary side resonant inductor current i_s is zero. At time t_b , the gate voltages v_{s1} and v_{s2} (only v_{s1} for the HBCLLC circuit) are applied. The primary side resonant inductor current i_p is negative, which means the current freewheels through the body diodes of S_1 and S_4 (only S_1 for the HBCLLC circuit), therefore S_1 and S_4 (only S_1 for

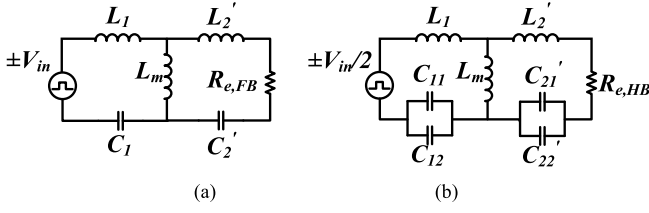


Fig. 4. Equivalent circuits of (a) FBCLLC and (b) HBCLLC converters in charging mode.

the HBCLLC circuit) will turn ON with ZVS at t_1 . Beyond t_b , power transfers from the primary side to the secondary side through the transformer and i_s is positive. Between t_b and t_c , i_p resonates and the magnetizing inductance current i_m keeps increasing almost linearly, since L_m is much larger than L_1 . When i_p meets i_m at t_c , previous resonance stops, and no power transfers to the secondary side, hence i_s becomes zero. The body diodes of S_6 and S_7 (only S_7 for the HBCLLC circuit) will turn OFF with ZCS naturally. The other half-cycle has a similar operating mode but with the opposite current direction.

2) *Gain Analysis:* The equivalent circuits of the FBCLLC and HBCLLC converters for charging mode are shown in Fig. 4(a) and (b), respectively. R_e , L_2' , and C_2' are the equivalent R_o , L_2 , and C_2 of the converters, respectively.

The general transfer function $H(s)$ of CLLC converters can be derived as follows:

$$H(s) = \frac{1}{n} \cdot \frac{R_e}{R_e + Z'_{L2} + Z'_{C2}} \cdot \frac{(R_e + Z'_{L2} + Z'_{C2}) \parallel Z_{Lm}}{Z_{L1} + Z_{C1} + (R_e + Z'_{L2} + Z'_{C2}) \parallel Z_{Lm}}. \quad (1)$$

The gain of the FBCLLC converter can be calculated as

$$G_{\text{CLLC}, G2V} = |H(s)| = \left| \frac{V_{\text{out}}}{V_{\text{in}}} \right| = \frac{1}{n} \cdot \frac{1}{\sqrt{a^2 + b^2}} \quad (2)$$

where

$$a = \frac{1}{h} + 1 - \frac{1}{h \cdot \omega^2} \quad (3)$$

$$b = \left(\frac{k}{h} + 1 + \frac{1}{g \cdot h} + \frac{1}{g} \right) \frac{Q}{\omega} - \left(\frac{k}{h} + 1 + k \right) Q \cdot \omega - \frac{Q}{g \cdot h \cdot \omega^3} \quad (4)$$

$$\begin{cases} h = \frac{L_m}{L_1}, k = \frac{L_2'}{L_1}, g = \frac{C_2'}{C_1}, \omega = \frac{\omega_s}{\omega_r} \\ \omega_r = \frac{1}{\sqrt{L_1 C_1}}, Q = \frac{\sqrt{L_1/C_1}}{R_e} \end{cases} \quad (5)$$

ω_r and ω_s are the resonant frequency and operating frequency, respectively. ω is the normalized frequency and Q is the quality factor. The first harmonic approximation is applied to calculate the equivalent load. For the FBCLLC circuit, the equivalent load, inductance, and capacitance can be calculated as

$$R_{e,FB} = (8n^2/\pi^2)R_o, \quad L_2' = n^2 L_2, \quad C_2' = C_2/n^2. \quad (6)$$

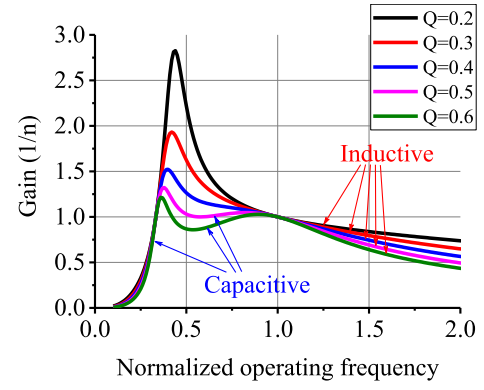


Fig. 5. Gain curves versus normalized frequency of CLLC converters at different loads.

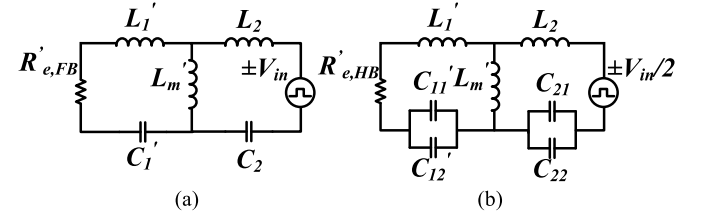


Fig. 6. Equivalent circuits of (a) FBCLLC and (b) HBCLLC converters in discharging mode.

Similarly, for the HBCLLC circuit, they are derived as follows:

$$R_{e,HB} = (2n^2/\pi^2)R_o, \quad L_2' = n^2 L_2, \quad C_1 = C_{11} + C_{12} \\ C_2' = C_{21} + C_{22}, \quad C_{21}' = C_{21}/n^2, \quad C_{22}' = C_{22}/n^2. \quad (7)$$

Fig. 5 shows the gain curves versus normalized frequency at different loads. The gain increases, but the slope of the curve decreases with a lower Q . To simplify the design, for both the FBCLLC and HBCLLC converters, k and g are set to be 1, which means $L_1 = L_2'$ and $C_1 = C_2'$, and h is set to be 4.

In the discharging mode, the equivalent circuits of the FBCLLC and HBCLLC converters are shown in Fig. 6(a) and (b), respectively.

For the FBCLLC circuit, the parameters will be derived as

$$R'_{e,FB} = (8/n^2\pi^2)R_o, \quad L_1' = L_1/n^2 \\ C_1' = n^2 C_1, \quad L_m' = \frac{L_m}{n^2}. \quad (8)$$

Similarly, for the HBCLLC circuit, they are derived as follows:

$$R'_{e,HB} = (2/n^2\pi^2)R_o, \quad L_1' = L_1/n^2, \quad C_1' = C_{11}' + C_{12}' \\ C_2 = C_{21} + C_{22}, \quad C_{11}' = n^2 C_{21}, \quad C_{12}' = n^2 C_{22} \\ L_m' = L_m/n^2. \quad (9)$$

The gain of the converter in the discharging mode is derived as follows:

$$G_{\text{CLLC}, V2G} = n \cdot \frac{1}{\sqrt{c^2 + d^2}} \quad (10)$$

where

$$c = \frac{1}{h'} + 1 - \frac{1}{h' \cdot \omega'^2} \quad (11)$$

$$d = \left(\frac{k'}{h'} + 1 + \frac{1}{g' \cdot h'} + \frac{1}{g'} \right) \frac{Q'}{\omega'} - \left(\frac{k'}{h'} + 1 + k' \right) \frac{Q' \cdot \omega'}{g' \cdot h' \cdot \omega'^3} \quad (12)$$

$$\begin{cases} h' = \frac{L'_m}{L_2}, k' = \frac{L'_1}{L_2}, g' = \frac{C'_1}{C_2}, \omega' = \frac{\omega_s}{\omega'_r} \\ \omega'_r = \frac{1}{\sqrt{L_2 C_2}}, Q' = \frac{\sqrt{L_2/C_2}}{R'_e} \end{cases} \quad (13)$$

With the same L and C values, the ω'_r , k' , g' , and h' of both the FBCLLC and HBCLLC circuits will maintain the same values as the ω_r , k , g , and h in charging mode, respectively. However, Q' will change since the equivalent load changes. Q' can be calculated as

$$Q' = n^2 R_o / R'_o Q. \quad (14)$$

3) *Soft-Switching Region*: Fig. 5 shows the inductive and capacitive resonate network regions of the CLLC circuit. The resonant network is inductive when the slope of the gain is negative. ZVS can be realized in the inductive region. To ensure the primary switches turn ON with ZVS, the magnetizing inductor current should be large enough to fully charge/discharge the output capacitors of the MOSFETs during the dead band time. The maximum value of L_m for an FBCLLC converter is derived in [17] as follows:

$$L_{m,FB} \leq \frac{t_{db}}{16 C_{oss} f_{s,max}}. \quad (15)$$

Similarly, for an HBCLLC converter, the L_m can be calculated as

$$L_{m,HB} \leq \frac{t_{db}}{8 C_{oss} f_{s,max}} \quad (16)$$

where t_{db} is the dead band time duration, C_{oss} is the output capacitance of the MOSFET, and $f_{s,max}$ is the maximum switching frequency.

B. Bidirectional DAB Converters

1) *Operating Principles*: Typical steady-state waveforms of an FBDAB converter [shown in Fig. 2(a)] with SPS control at heavy load conditions are shown in Fig. 7(a). It has been assumed that $V_{in}/n < V_o$ and $i_L(t_a) < 0$. T_s is the switching period, and $T_s = 1/f_s$, where f_s is the switching frequency. The phase shift between the two bridges t_{shift} is $DT_s/2$, and D is from 0 to 0.5. v_1 is the voltage on the secondary side of the transformer, and v_2 is the input voltage of the rectifier. v_L is the voltage across the inductor, L , therefore, $v_L = v_1 - v_2$. i_L is the inductor current. Prior to t_a , Q_2 , Q_3 , Q_5 , and Q_8 are ON, and Q_1 , Q_4 , Q_6 and Q_7 are OFF. At t_a , Q_1 and Q_4 turn ON, Q_2 and Q_3 turn OFF. t_{shift} later, Q_6 and Q_7 turn ON, Q_5 and Q_8 turn OFF. i_L is negative at t_a and positive at t_b , which means Q_1 , Q_4 , Q_6 , and Q_7 turn ON with ZVS. However, in light load conditions, i_L may be positive at t_a and negative at t_b , as shown in Fig. 7(b), which means ZVS cannot be achieved. The detailed analysis can be found in [20].

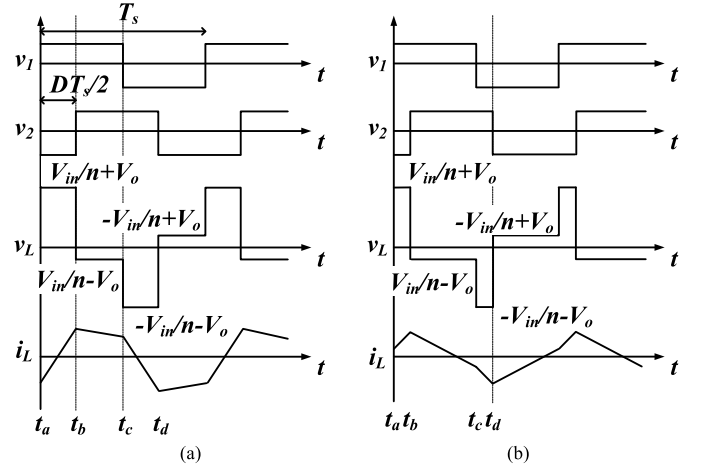


Fig. 7. Typical waveforms of an FBDAB converter for $V_{in}/n < V_o$ and $i_L(t_a) < 0$ in (a) heavy load condition and (b) light load condition.

The other half-cycle has similar operating modes but with the opposite current direction. The HBDBAB circuit has similar waveforms with half-inductor voltage.

2) *Gain Analysis*: Based on the analysis in [20], the gain of the FBDAB converter can be calculated as

$$G_{FBDAB,G2V} = \frac{V_{out}}{V_{in}} = \frac{1}{n} \cdot \frac{R_o}{2f_s L} D(1-D). \quad (17)$$

The output power is derived as

$$P_{FBDAB,G2V} = \frac{V_{in} V_{out}}{2nf_s L} D(1-D). \quad (18)$$

Similarly, the gain and output power of the HBDBAB can be derived, respectively, as follows:

$$G_{HBDBAB,G2V} = \frac{1}{n} \cdot \frac{R_o}{8f_s L} D(1-D) \quad (19)$$

$$P_{HBDBAB,G2V} = \frac{V_{in} V_{out}}{8nf_s L} D(1-D). \quad (20)$$

In the discharging mode, the gain and output power of the FBDAB can be derived, respectively, as follows:

$$G_{FBDAB,V2G} = \frac{1}{n} \cdot \frac{R'_o}{2f_s L} D(1-D) \quad (21)$$

$$P_{FBDAB,V2G} = \frac{V_{in} V_{out}}{2nf_s L} D(1-D). \quad (22)$$

For the HBDBAB circuit

$$G_{HBDBAB,V2G} = \frac{1}{n} \cdot \frac{R'_o}{8f_s L} D(1-D) \quad (23)$$

$$P_{HBDBAB,V2G} = \frac{V_{in} V_{out}}{8nf_s L} D(1-D). \quad (24)$$

3) *Soft-Switching Region*: According to the analysis, to ensure fully ZVS operation, the inductor current is required to be negative at t_a ($i_L(t_a) \leq 0$) and positive at t_b ($i_L(t_b) \geq 0$). The requirements of phase shift D can be calculated as follows [20]:

$$\begin{cases} D \geq \frac{G_{DAB}-1}{2G_{DAB}} & \text{if } G_{DAB} \geq 1 \\ D \geq \frac{1-G_{DAB}}{2} & \text{if } G_{DAB} \leq 1. \end{cases} \quad (25)$$

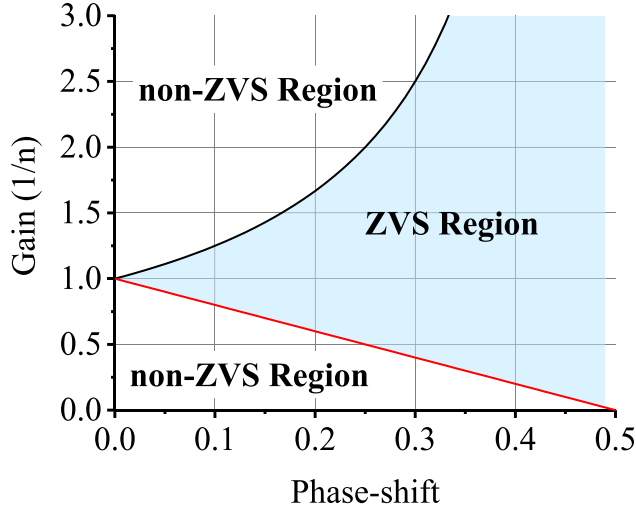


Fig. 8. ZVS region of DAB converters.

Equation (25) can be applied to the HBDAB circuit as well. Based on (25), the ZVS region of both the FBDAB and HBDAB converters is shown in Fig. 8. The converters can always operate under ZVS condition when the gain is $1/n$. Otherwise, by lowering the phase shift, the ZVS region of the converter decreases. Therefore, in light conditions, since the phase shift must be smaller to maintain the gain, the converters may lose soft switching, which will cause large switching losses.

III. DESIGN METHODOLOGY

In the charging mode, the input voltages of all the four topologies are 500 V, and the output voltage range is 200–420 V. For the discharging mode, the input voltage range is 350–420 V, and the output voltage is 500 V. The nominal operating frequency for both the charging and discharging modes is 170 kHz.

A. Bidirectional CLLC Converters

For the CLLC converters, the transformer turns ratio n is set to be 1.5 in order to step down the input voltage and simplify the control. The G2V gain, therefore, can be designed to be $0.6/n$ – $1.3/n$, and the V2G gain range should be wider than $0.79n$ – $0.96n$. Furthermore, the curve of the two gains should decrease monotonically in the designed zone for linear control. From Fig. 6, a larger Q can give a narrower operating frequency range, whereas a smaller Q can guarantee the gain and the monotony of the gain. In this design, $Q = 0.4$ is chosen for full load condition ($R_o = 176\Omega$) in charging mode and the resonant frequency f_r is designed to be 170 kHz. For a simplified design, C_1 is equal to C_2' , and L_1 is equal to L_2' . The relation between Q and L_m is given in [28]. A large L_m reduces the peak current, which lowers conduction loss, but it will also reduce the gain of the circuit. Furthermore, from (15) and (16), a larger L_m may cause hard switching of the power MOSFETs. Therefore, considering the tradeoff between gain and conduction loss, L_m is chosen to be four times larger than L_1 . Hence, L_1 , L_2 , L_m , C_1 , and C_2 for the FBCLLC circuit can be calculated from (5) and (6),

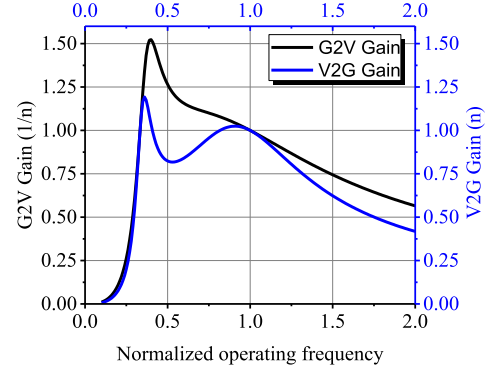


Fig. 9. Gains of the CLLC circuits in charging and discharging modes.

which will lead to

$$\begin{cases} L_1 = 120.2 \mu\text{H}, L_2 = 53.4 \mu\text{H} \\ C_1 = 7.3 \text{ nF}, C_2 = 16.4 \text{ nF} \\ L_m = 480.8 \mu\text{H}. \end{cases} \quad (26)$$

Similarly, the parameters for the HBCLLC circuit can be calculated from (5) and (7) as follows:

$$\begin{cases} L_1 = 30.1 \mu\text{H}, L_2 = 13.4 \mu\text{H} \\ C_{11} = C_{12} = 14.6 \text{ nF}, C_{21} = C_{22} = 32.8 \text{ nF} \\ L_m = 120.4 \mu\text{H}. \end{cases} \quad (27)$$

Fig. 9 illustrates the gains of the charging and discharging modes for both the FBCLLC and HBCLLC circuits with the parameters calculated above, which satisfy the design requirements.

B. Bidirectional DAB Converters

Equations (17) and (21) show that the transformer turns ratio n has the same effect on the gains of DAB circuit in both charging and discharging modes. For example, if $n > 1$, the transformer will help step down the voltage in charging mode, however, in discharging mode, the transformer will still step down the voltage. In this design, the transformer turns ratio n is set to be 1.5 to balance the control pressure in both power flows and fully use the ZVS region shown in Fig. 8. The G2V gain, hence, can be designed to be $0.6/n$ – $1.3/n$, and the V2G gain range should be wider than $1.7/n$ – $2.2/n$. The operating frequency f_s of this circuit is 170 kHz. To reduce the reactive power [26] and make the gain more linear, the maximum phase shift $D_{\max}$ is set to be 0.45 for the worst case. Since the system cannot operate with full power when $V_o = 200$ V for battery charging, the worst case happens in discharging mode with $V_{in} = 350$ V, $V_o = 500$ V, and $P = 1$ kW.

For the FBDAB circuit, the inductance L , therefore, can be calculated from (22) as

$$L_{\text{FBDAB}} = \frac{V_{in} V_{out}}{2n f_s P} D_{\max} (1 - D_{\max}) = 84.9 \mu\text{H}. \quad (28)$$

Similarly, for the HBDAB circuit, the inductance L can be calculated from (24) as

$$L_{\text{HBDAB}} = \frac{V_{in} V_{out}}{8n f_s P} D_{\max} (1 - D_{\max}) = 21.2 \mu\text{H}. \quad (29)$$

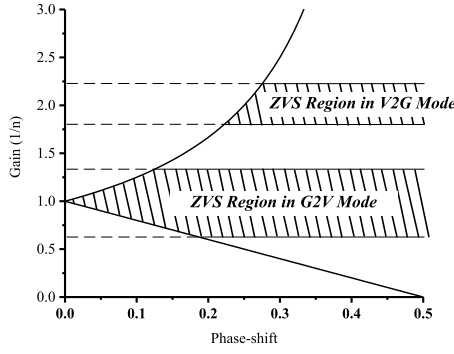


Fig. 10. Full ZVS regions of the DAB converters in both charging and discharging modes.

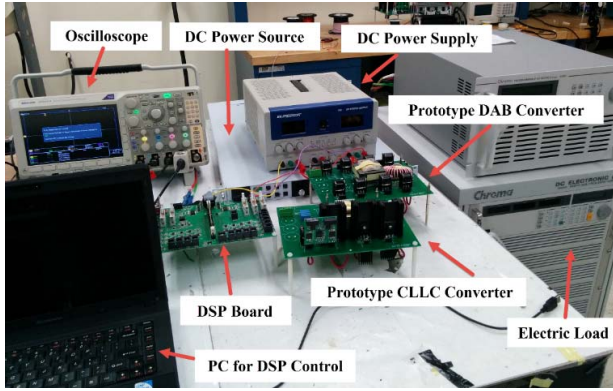


Fig. 11. Experimental setup to test the prototypes.

TABLE I
COMPONENTS OF CLLC CONVERTERS

FBCLLC		HBCLLC	
Component	Product/Value	Component	Product/Value
$S_1 - S_8$	C2M0080120D	S_1, S_2, S_7, S_8	C2M0080120D
n	18:12	n	12:8
L_m	522 μ H	L_m	131 μ H
L_1	131 μ H	L_1	32 μ H
L_2	60 μ H	L_2	15 μ H
C_1	7.8 nF	C_{11}, C_{12}	14.2 nF
C_2	17.0 nF	C_{21}, C_{22}	33.0 nF

From (25), the minimum phase shift can be calculated for each gain. In this design, for both the FBDAB and HBDAB converters, the full ZVS regions in both the charging and discharging modes are shown in the shadowed areas in Fig. 10. At light loads, the phase shift is reduced to maintain the gain; however, in this condition, the converters lose ZVS.

IV. EXPERIMENTAL RESULTS

Fig. 11 illustrates the experimental setup for testing the prototypes. The components of the CLLC and DAB circuits are listed in Tables I and II, respectively.

The switches are SiC power MOSFETs, which have high voltage and current ratings with low output capacitance. A TMS320F28335 DSP-based digital control platform is used to generate the control signal for the circuits. Since the magnetizing inductance L_m of the transformer in the FBCLLC circuit

TABLE II
COMPONENTS OF DAB CONVERTERS

FBCLLC		HBCLLC	
Component	Product/Value	Component	Product/Value
$Q_1 - Q_8$	C2M0080120D	Q_1, Q_2, Q_7, Q_8	C2M0080120D
n	12:8	n	12:8
L	90 μ H	L	22 μ H
		$C_{11}, C_{12}, C_{21}, C_{22}$	30 μ F

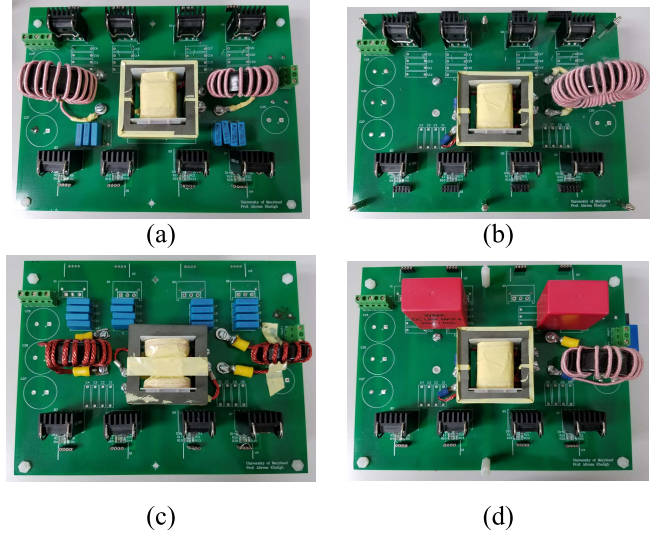


Fig. 12. Different dc-dc converter prototypes. (a) FBCLLC. (b) FBDAB. (c) HBCLLC. (d) HBDAB.

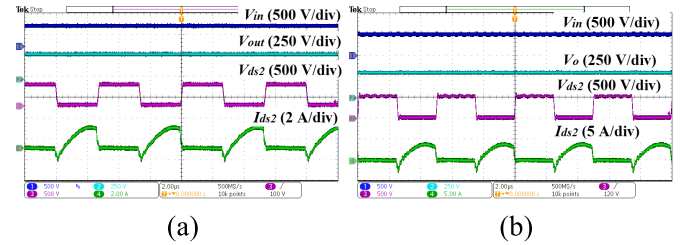


Fig. 13. Experimental waveforms of (a) FBCLLC circuit and (b) HBCLLC circuit at load = 800 W, $V_o = 300$ V, and $f_s = 200$ kHz.

is higher than that in the HBCLLC circuit, the transformer in the FBCLLC circuit requires more number of turns to ensure a higher magnetizing inductance.

Fig. 12 shows the four prototypes. Since the four converters have similar topologies and for simplicity as well as rapid prototyping, the converters are tested based on one general Printed Circuit Board platform. The film capacitors in CLLC converters are connected in parallel to reduce Equivalent Series Resistance. Other two-bridge capacitors are connected at the bottom of the HBDAB board.

Fig. 13(a) and (b) illustrates the experimental waveforms of the FBCLLC and the HBCLLC circuits in the charging mode. The load is 800 W, $V_o = 300$ V, and $f_s = 200$ kHz. It is clear that for each of the circuits, the switches turn ON with ZVS. The circuits have similar experimental results in the discharging mode since the structures of the circuits are symmetrical.

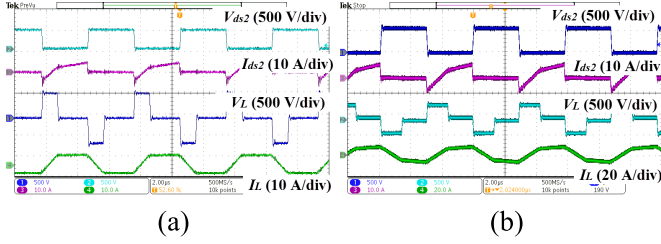


Fig. 14. Experimental waveforms of (a) FBDAB circuit and (b) HBDAB circuit at load = 800 W, $V_o = 300$ V, $f_s = 170$ kHz, and $D = 0.4$.

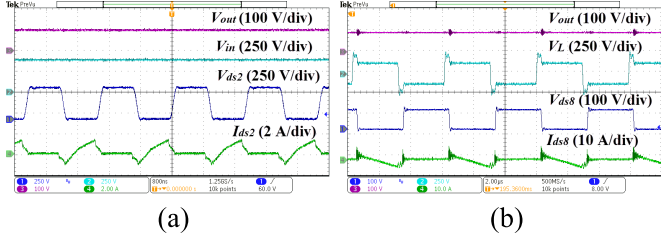


Fig. 15. Experimental waveforms of (a) HBCLLC circuit and (b) HBDAB circuit at load = 50 W and $V_o = 100$ V.

TABLE III
COMPARISON OF THE CONVERTERS

	FBCLLC	HBCLLC	FBDAB	HBDAB
Soft-switching Region	Full load	Full load	Partial load	Partial load
Control Modulation	Frequency	Frequency	Phase-shift	Phase-shift
Design Complexity	High	High	Low	Low

Fig. 14(a) and (b) shows the FBDAB and HBDAB circuit experimental waveforms in the charging mode at phase shift $D = 0.4$, respectively. The load is 800 W, $V_o = 300$ V, and $f_s = 170$ kHz. In this case, $V_o < V_{in}/n$. The waveforms show that the switches turn ON with ZVS. Similar experimental results are obtained in the discharging mode.

Fig. 15 shows an example at light load condition. The HBCLLC and HBDAB converters operate at 100 V output voltage and 50 W load conditions. Since the CLLC converter is controlled through frequency modulation and the DAB converter is controlled through phase shift modulation, the operating frequency is 500 kHz for the HBCLLC converter and remains 170 kHz for the HBDAB converter. The phase shift D of the HBCLLC converter is set to be 0.1 for this light load condition. It is clear that the switch of the HBCLLC converter still operates with ZVS, whereas the secondary side switch of the HBDAB converter loses ZVS. The experimental results verify the previous analysis.

Fig. 16(a) and (b) shows the efficiencies of the four circuits in the charging and discharging modes, respectively. The output voltage is 300 V for the charging mode and 500 V for the discharging mode.

V. EXPERIMENTAL RESULTS ANALYSIS

The summaries of the comparisons are presented in Table III and Fig. 17.

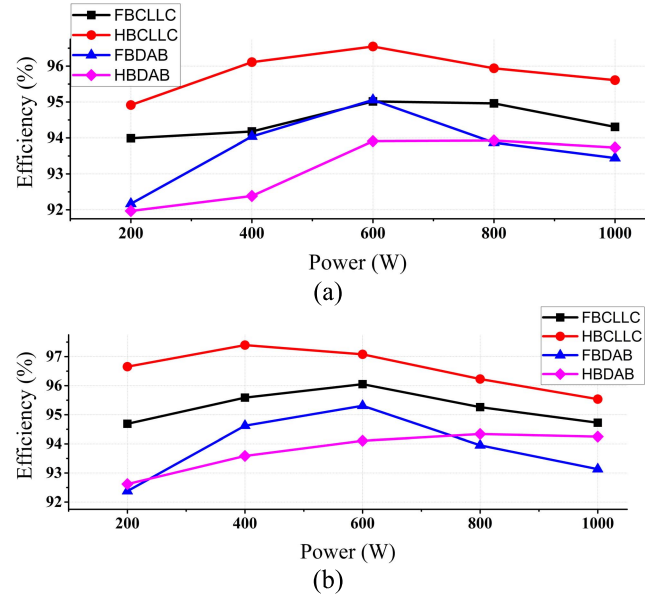


Fig. 16. Measured efficiency of the four circuits in (a) charging and (b) discharging modes.

A. Efficiency

From the experimental results, all the circuits achieve high efficiency. The highest efficiencies in the charging mode are 95.0% and 96.5% for the FBCLLC and the HBCLLC circuits, respectively, and 96.1% and 97.4% in the discharging mode. The highest efficiencies are achieved when the circuits operate close to the resonant frequency. When the operating frequency is much smaller or larger than the resonant frequency, the circulating loss will increase and reduce the efficiency. For the FBDAB and HBDAB circuits, the highest efficiencies are 95.1% and 93.9% in the charging mode, and 95.3% and 94.3% in the discharging mode, respectively. The efficiencies of the CLLC circuits are higher than those of the DAB circuits, since the switches of the DAB circuits lose ZVS at light load conditions, and the SPS control strategy creates relatively large reactive power in the circuits, which also reduces the efficiency.

In addition, compared with the full-bridge structure, the half-bridge structure helps the circuits improves the overall efficiency, due to the reduction of the total number of switches and the corresponding driving circuits. An extra benefit of the half-bridge structure of the resonant converters is that the two capacitors in each leg can automatically reduce flux imbalance for the transformer.

B. Soft-Switching Region

From the analysis in Section II, the soft-switching region of the CLLC circuits are much wider than that of the DAB circuits especially at light load conditions. With an appropriate design, both the FBCLLC and HBCLLC converters can achieve full soft switching in the entire load range. However, for DAB converters, Figs. 10 and 15 show that the converters lose ZVS at light load conditions. Therefore, it is difficult to maintain soft switching of all the switches for a wide load range. Furthermore, for CLLC circuits, lighter loads provide

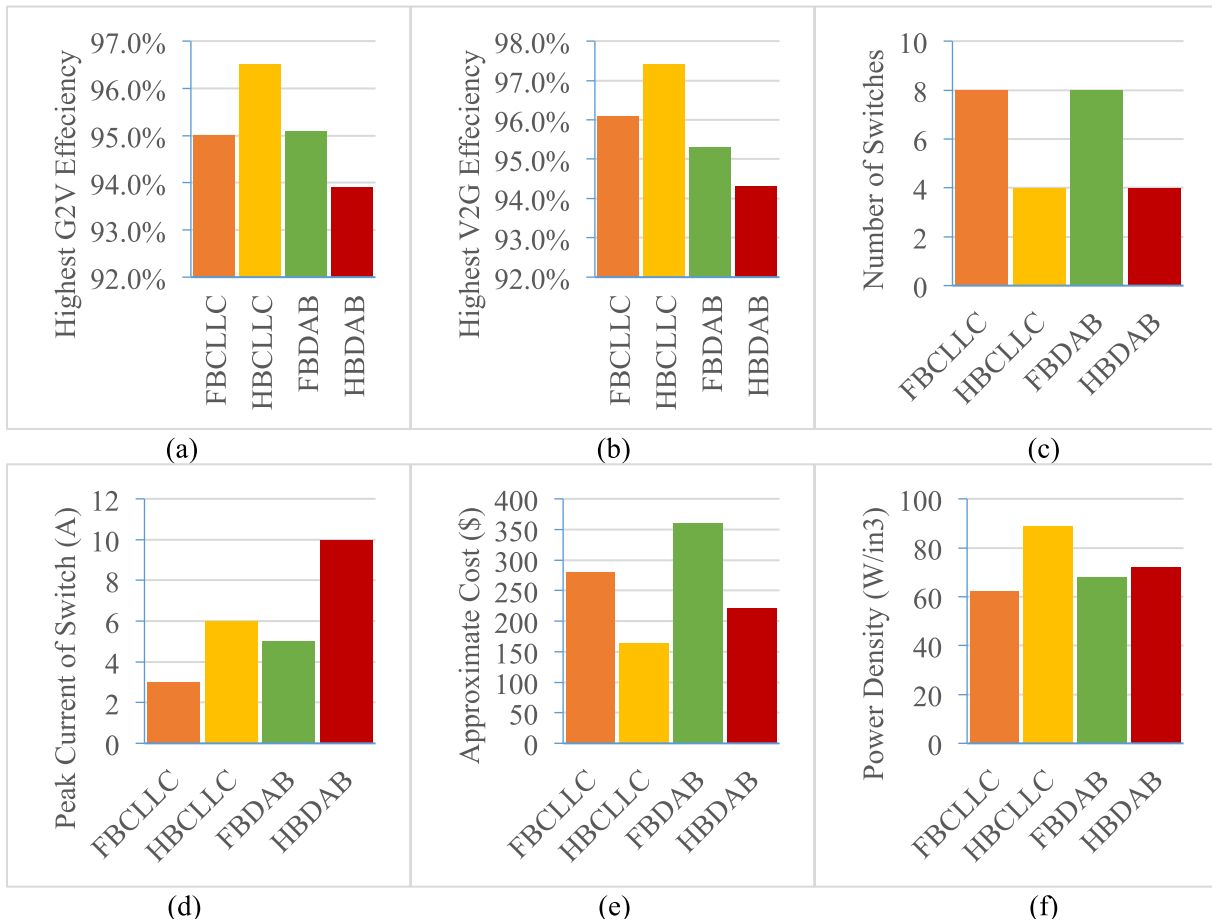


Fig. 17. Comparison for (a) highest G2V efficiency, (b) highest V2G efficiency, (c) number of switches, (d) peak current of switch, (d) approximate cost of each converter, and (f) power density.

wider soft-switching regions, whereas for DAB circuits, it is the opposite.

C. Design and Control Complexity

According to the analysis in Section II, if the CLLC circuits operate close to their resonant frequencies, the gains will be about $1/n$, independent of the loads. Therefore, if the operating frequency can be adjusted at the vicinity of the resonant frequency, both the FBCLLC and HBCLLC circuits are perfect for constant output voltage, variable load conditions. However, the gains of the CLLC circuits are nonlinear with the loads. Furthermore, depending on the value of Q and particularly at small Q values, the required operating frequency range will be very large. Hence, in the case of a wide output voltage range, the value of Q should be carefully chosen to limit the operating frequency range.

In the case of the DAB circuits, the gains are proportional to the loads, and almost linearly related to the phase shift when $D \leq 0.45$. Therefore, comparing the DAB converters with the CLLC converters, the design and control complexity for the DAB converters is less.

D. Component Selection

For the CLLC converter, the output current is sinusoidal, whereas for the DAB converter, the output current has more

high-order harmonic ripples than that of the CLLC converter. Therefore, an extra current filter may be needed for the DAB circuits.

Comparing the full-bridge and the half-bridge structures, under the same input and output conditions, in the half-bridge structure, both primary side and secondary side currents are twice larger than those in the full-bridge structure. Therefore, the switches, resonant components, and transformer of the half-bridge converters are under about twice larger current stress than those of the full-bridge converters.

E. Power Density

It is clear that, by reducing the number of switches, the half-bridge structure has benefits in terms of size and weight, since the number of corresponding driving circuits and cooling systems can be reduced at the same time. Furthermore, with the same gain, the half-bridge structure requires less inductance than the full-bridge structure. Therefore, the size of the converters with a half-bridge structure should be smaller than that with a full-bridge structure. Comparing the two half-bridge converters and the HBCLLC and HBDAB converters, although the HBCLLC converter requires one more inductor, the bridge capacitances of the HBCLLC converter are much smaller than those of the HBDAB converter, since they are also used as resonant capacitors. Therefore, in this comparison, the

size and weight of the HBCLLC circuit is less than that of the other circuits, which means the power density of the HBCLLC circuit is the highest. Furthermore, the SiC MOSFETs and their gate drivers cost the most in the converters, and reducing the number of switches will reduce the cost of the converters. Therefore, the half-bridge converters cost less than the full-bridge converters.

VI. CONCLUSION

This paper provides a comprehensive analysis and comparison among the FBCLLC, HBCLLC, FBDAB, and HBDAB converters for EV battery charging applications. A new general gain expression of the HBCLLC converter is derived in this paper. The practicability and performance of the four dc–dc converters for bidirectional EV charging systems are discussed. The converters are designed with a 1 kW power rating. All the converters can achieve high efficiency and bidirectional power flow. The topologies, operating principles, and design methodologies of the converters are discussed, and the performances of the converters are compared. Although the current stress on the HBCLLC converter is relatively high, the power density of the HBCLLC converter is the highest and the cost is relatively low. Soft switching can be achieved in full load range for the HBCLLC converter, and in addition, if the input voltage could be adjustable, the HBCLLC converter can be operated at the resonant frequency, which is with the highest efficiency. Therefore, for bidirectional wide load EV charging systems, the CLLC converters are slightly better than the DAB converters. Considering the efficiency, size, and cost, the HBCLLC converter will be the most suitable choice at 1 kW.

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